
Trigonometric Identities and Equations — Illegal Cancellation and Lost Solutions

Cancelling is the fastest way to lose a solution you were never told about. Three rules cover almost every case.

- **Never divide both sides by an expression containing the variable.** Move everything to one side and *factor* instead: the zero-product property then keeps every branch.
- **Only factors of a product cancel.** A term of a sum may never be cancelled against a denominator.
- **Squaring both sides can create solutions that are not there.** Every candidate must be checked in the *original* equation.
- In each problem: name the error in words first, then show corrected work. Unless told otherwise, solve on $0^\circ \leq \theta < 360^\circ$.

1. Find and fix. Solve $2 \sin \theta \cos \theta = \sin \theta$ on $0^\circ \leq \theta < 360^\circ$.

Rafi divides both sides by $\sin \theta$, gets $\cos \theta = \frac{1}{2}$, and reports $\theta = 60^\circ, 300^\circ$.

- (a) Name what Rafi's division assumed without saying so.
(b) Solve the equation correctly and list every solution.

Answer: _____

2. Find and fix. Simplify $\frac{\cos^2 x - 1}{\cos x - 1}$.

Bea cancels the $\cos^2 x$ against the $\cos x$ and the 1 against the 1, and writes $\cos x$.

(a) Simplify the expression correctly by factoring the numerator.

(b) Show Bea's form is wrong by evaluating both at a value where $\cos x = 0.5$, and give the correct value there.

Answer: _____

3. Find and fix. Solve $2 \cos^2 \theta = \cos \theta$ on $0^\circ \leq \theta < 360^\circ$.

Nils divides both sides by $\cos \theta$ and reports $\theta = 60^\circ, 300^\circ$.

(a) Write the equation in factored form with 0 on one side.

(b) List every solution, and say which ones Nils's division threw away.

Answer: _____

4. Find and fix. A student simplifies $\frac{\sin^2 x + \cos x}{\cos x}$ to $\sin^2 x + 1$.

- (a) Explain why the $\cos x$ in the numerator cannot be cancelled against the denominator.
- (b) Test both forms at a value where $\sin^2 x = 0.75$ and $\cos x = 0.5$. Give the correct value there.

Answer: _____

5. Find and fix. Solve $4 \sin^2 \theta = 1$ on $0^\circ \leq \theta < 360^\circ$.

Tam takes the square root of both sides, writes $\sin \theta = \frac{1}{2}$, and reports $\theta = 30^\circ, 150^\circ$.

- (a) Say what taking a square root leaves out.
- (b) List every solution on the interval.

Answer: _____

6. Find and fix. Solve $\sin \theta - \cos \theta = 1$ on $0^\circ \leq \theta < 360^\circ$.

Ola squares both sides, simplifies to $\sin \theta \cos \theta = 0$, and reports all four of $\theta = 0^\circ, 90^\circ, 180^\circ, 270^\circ$.

- (a) Ola's algebra is correct but her answer is not. Say what squaring did to the problem.
- (b) Test each candidate in the original equation and list the solutions that actually work.

Answer: _____

7. Find and fix. Solve $\cos 2\theta = \cos \theta$ on $0^\circ \leq \theta < 360^\circ$.

Yuki "cancels the cosine from both sides" to get $2\theta = \theta$, and reports $\theta = 0^\circ$ only.

- (a) Explain why equal cosine values do not force equal angles.
- (b) Use a double-angle identity to write the equation with 0 on one side, factor it, and list every solution.

Answer: _____

8. Explain.

- (a) Explain why dividing both sides of an equation by a trigonometric expression can lose solutions, and state what the division silently assumes.
- (b) Explain why a term of a sum may never be cancelled against a denominator, even though a factor of a product may. Give a numerical example that makes the difference visible.
- (c) Explain why squaring both sides forces you to check every candidate, and contrast that with what dividing does. One operation adds solutions and the other removes them — say which is which and why.